

Joint Optimal Pro Application

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This tool is intended for:

Full toolset for creating and orienting skeletons

most of these tools work not only with bones, but in general with any objects

Including:

- several ways to create bones
- an unique method of adding bones to a nonlinear chain
- tools for quick parenting
- a set of tools for alignment, smoothing and bone distribution
- a set of tools for selecting bones, including an unique tool for selection the shortest path between two bones
- a set of tools for orienting bones - automatic and manual
- tool for painting bones in various colors
- a set of tools for quick naming, searching for matching names, for removing namespaces, etc.

Installation:

- Unpack the .zip file to any place on the disk where it is convenient
- Open Maya.
- Drag and drop file:

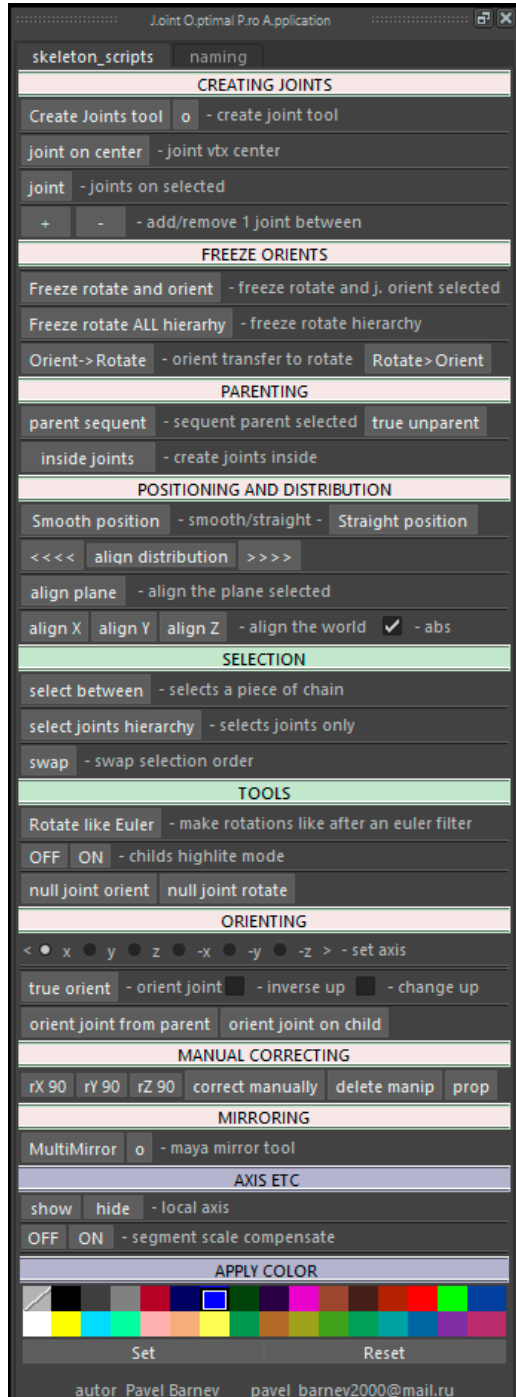
`drag_and_drop_THIS_FILE_on_maya_window_for_install_Joint_Optimal_Pro_Application.mel`

in open MAYA window. An icon will appear in the Custom tab of the Shelf Editor.

general approach:

- first we create bones in the right places,
- align them
- and only then orient in the right direction

general form:



Creating joints toolset

Freezing and converting
Rotate and Joint Orient

Parenting selection

Smoothing Distribution
and Aligning joint positions

Selection toolset

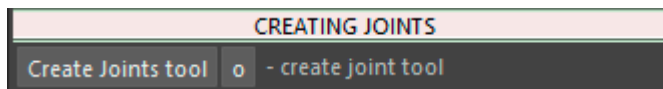
Orienting Joints toolset

Manual correcting joins rotation

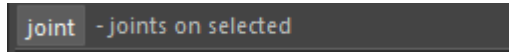
Set color to selection

Details:

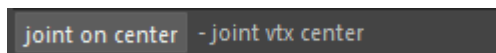
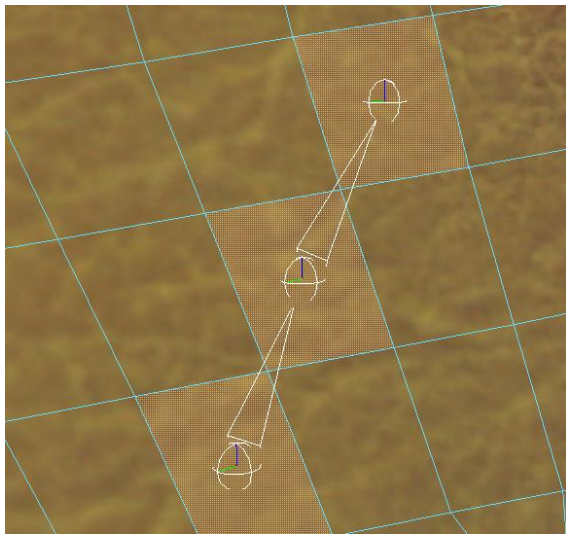
Creating Joints part:



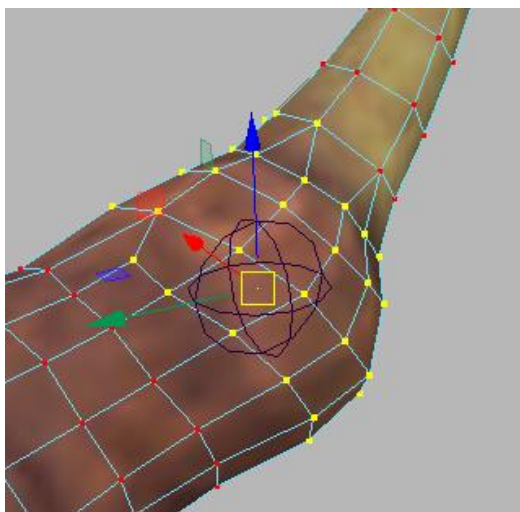
Maya standart Crate Joints tool and options (recommmed use with Snap to projected center option)

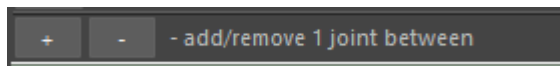


Creating joints on selected objects, vertexes , center of edges or center of poly by selection order



Creating single joint on center selection

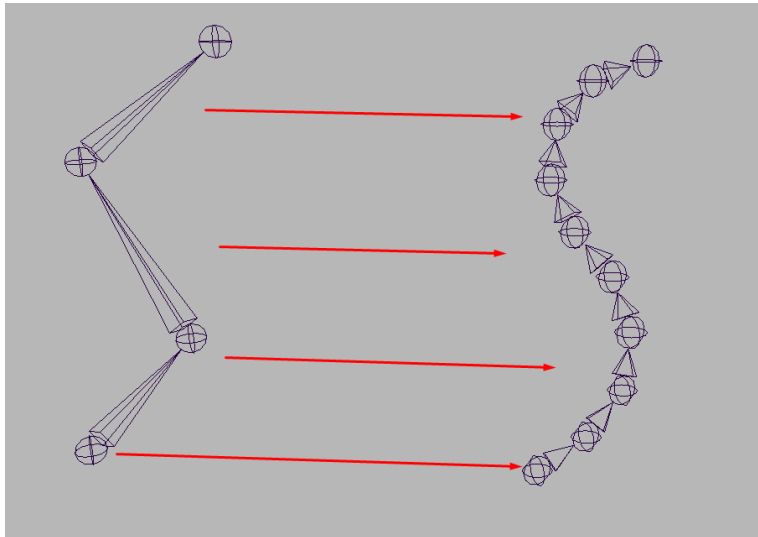




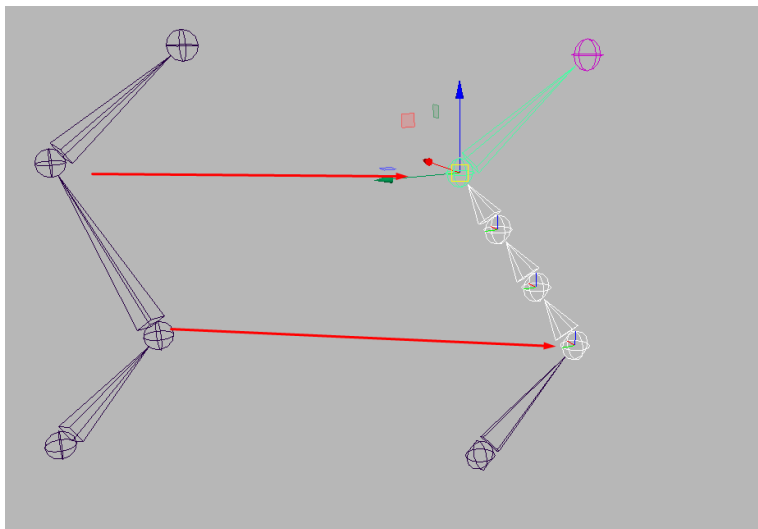
Add / remove 1 joint in selected joints chain can work with a nonlinear chain

Can work with a part of chain

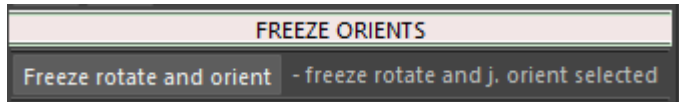
Result of adding 7 bones to selected 4 joint chain



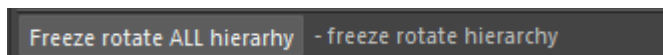
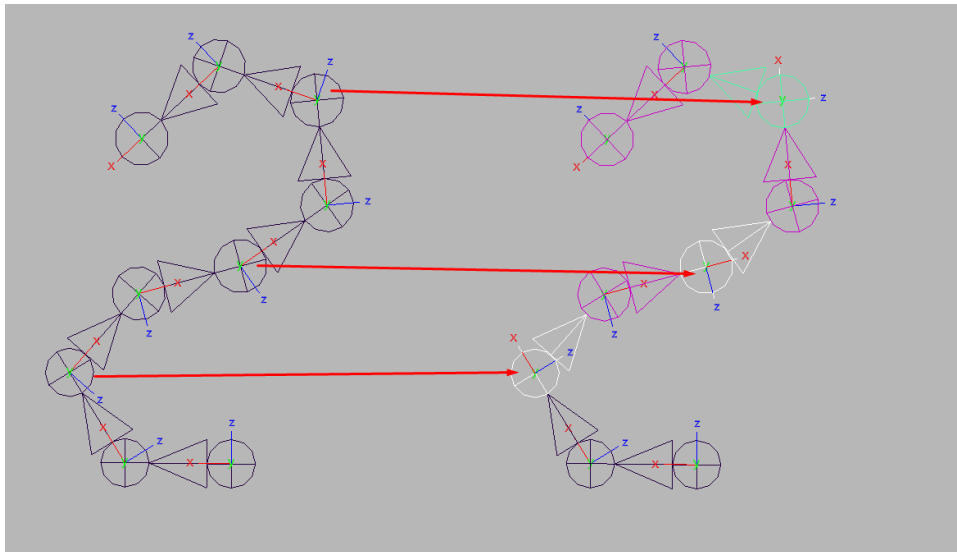
Can work with a part of chain



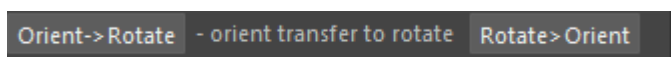
FREEZE ORIENTS part:



Freeze rotate and orient on selected joints leaving the orientation of children

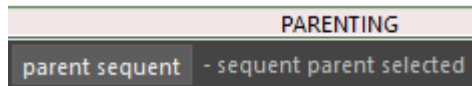


nullifies the rotation of the bones in the hierarchy and transform them into a Joint Orient

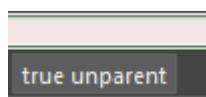
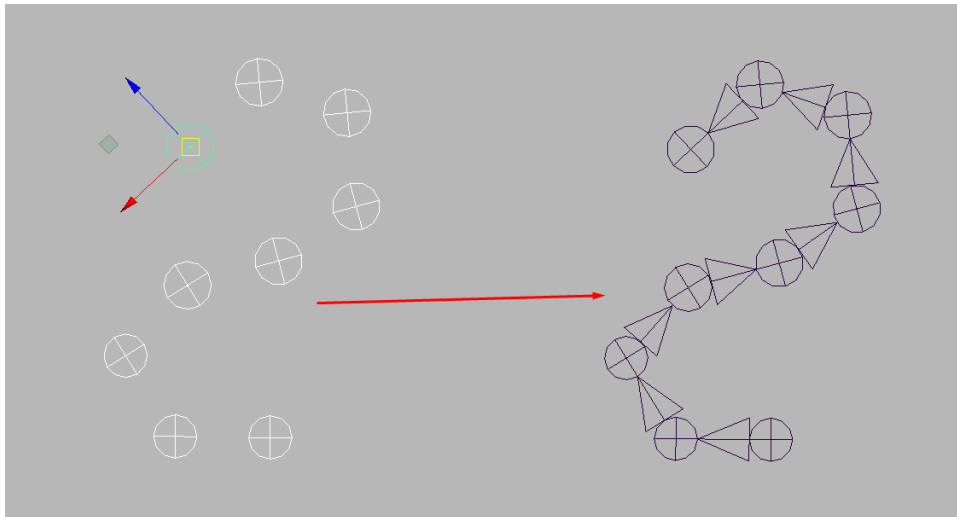


Transfer Orient to Rotate or Rotate to Orient on selected joints, excluding children

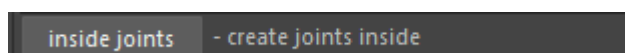
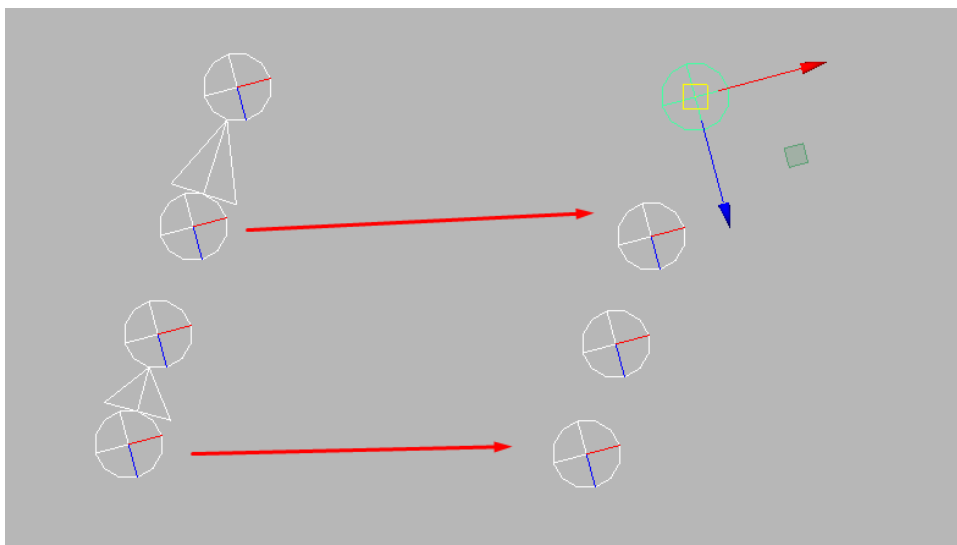
PARENTING part



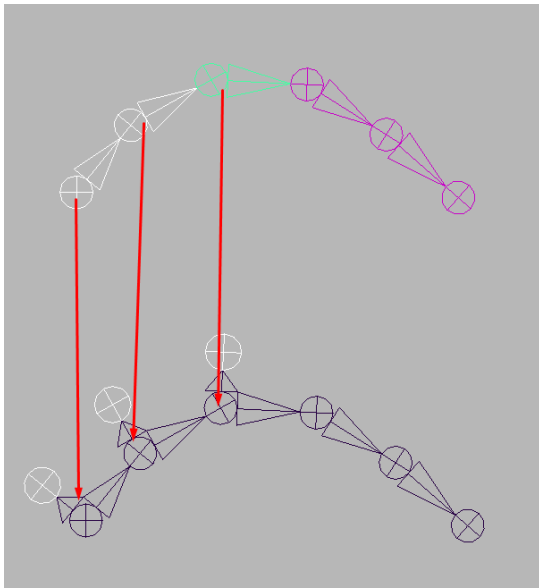
Sequentially parenting selected joints



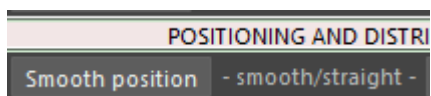
Unparent selected joint chains with saving selection on top hierarchy



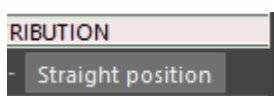
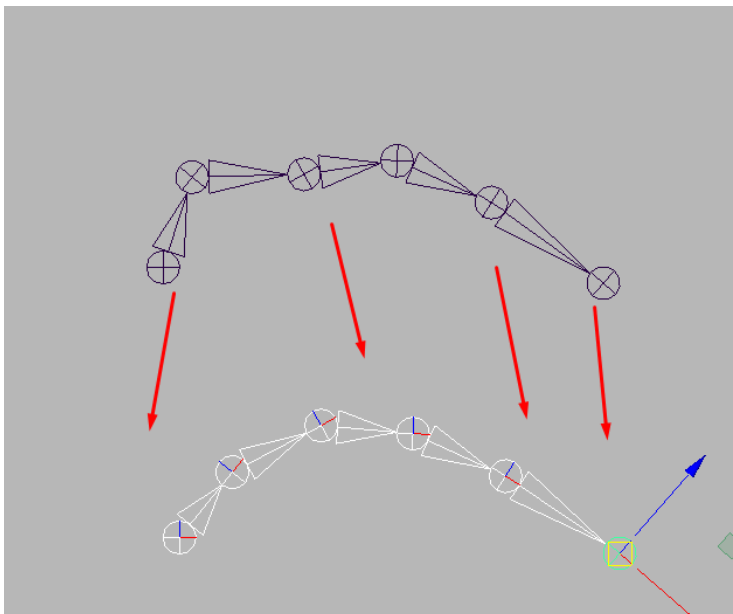
Creating joints inside selected



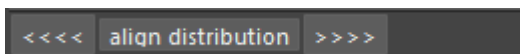
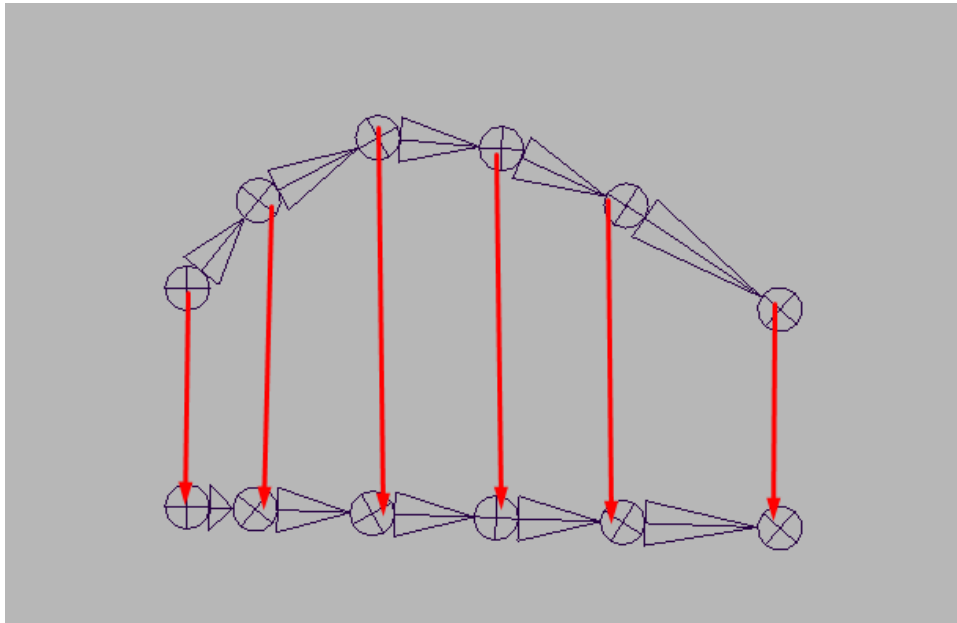
POSITIONING AND DISTRIBUTION part



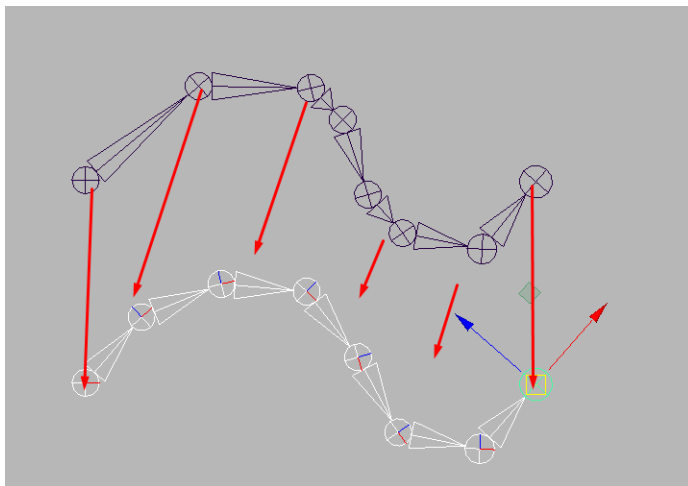
Smoothing position of selected joints



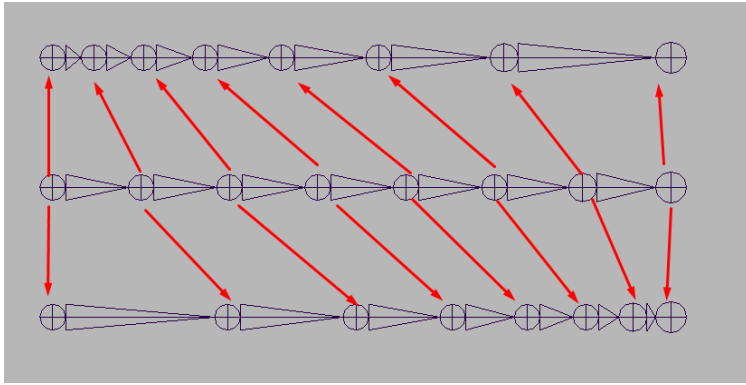
Straight position of selected joints



Align distribution of selected joints with saving total form of chain

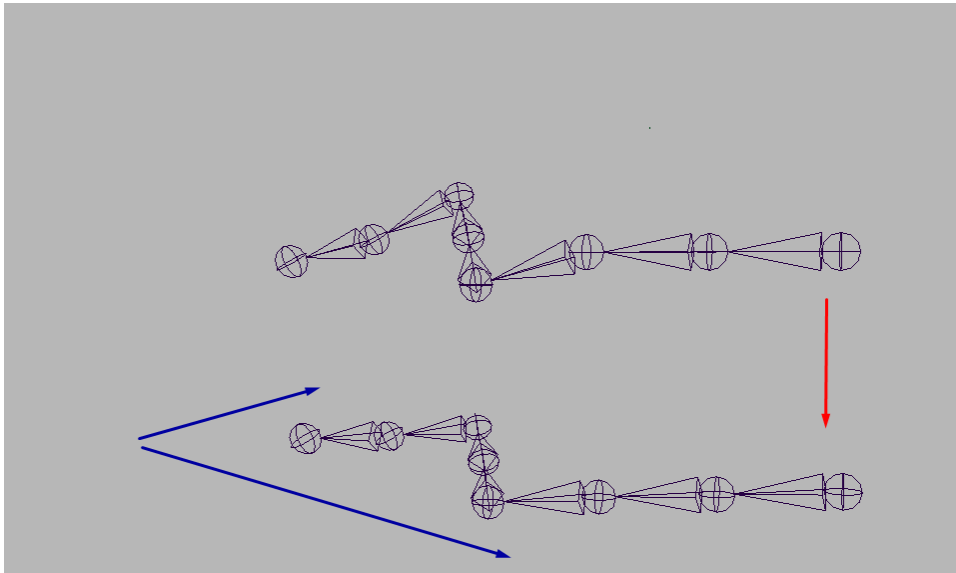


With Right or Left side



align plane - align the plane selected

Align selected to calculated common plane

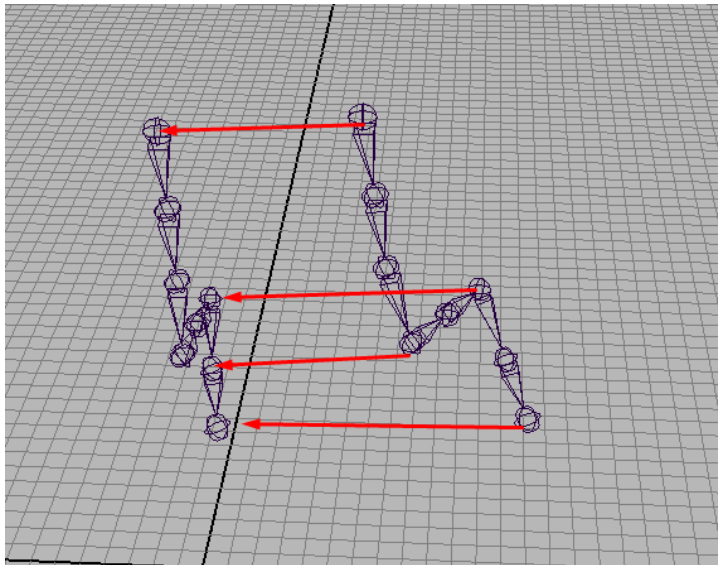


align X align Y align Z - align the world - abs

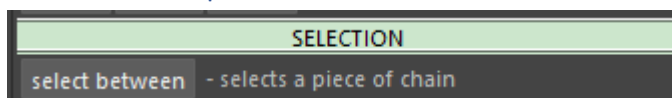
Align selected to world Axis plane

If ABS is OFF – the world axis passes through the first selected

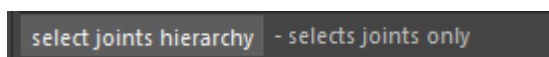
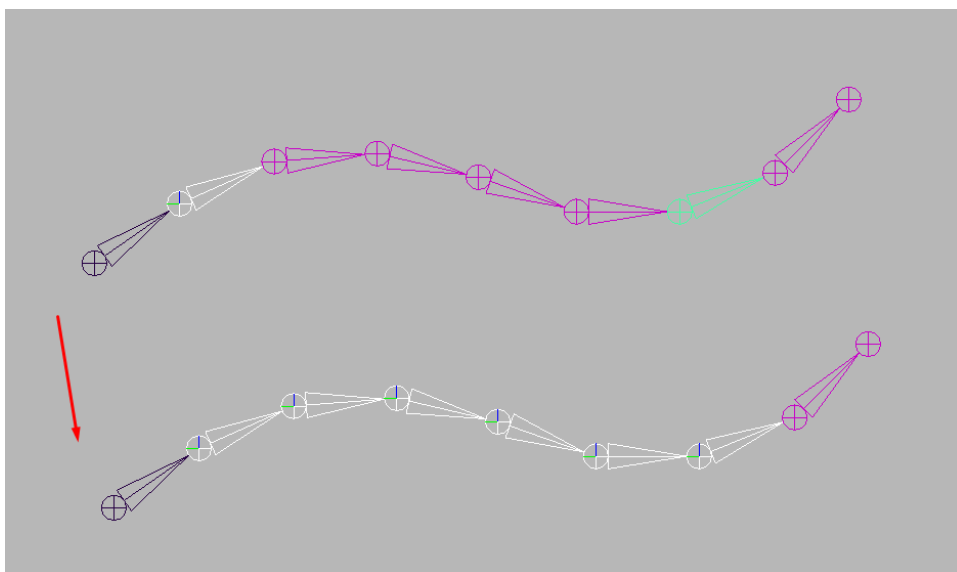
If ABS is ON – the world axis passes through 0 0 0 coordinates



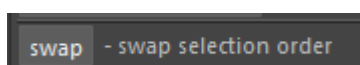
SELECTION part:



Select shortcut between 2 joints (selected by hierarchy order)

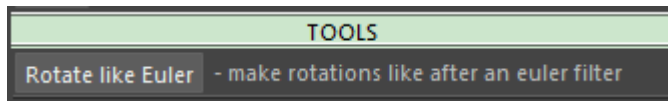


Select by hierarchy joints only

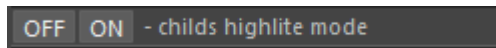


Revers selection order

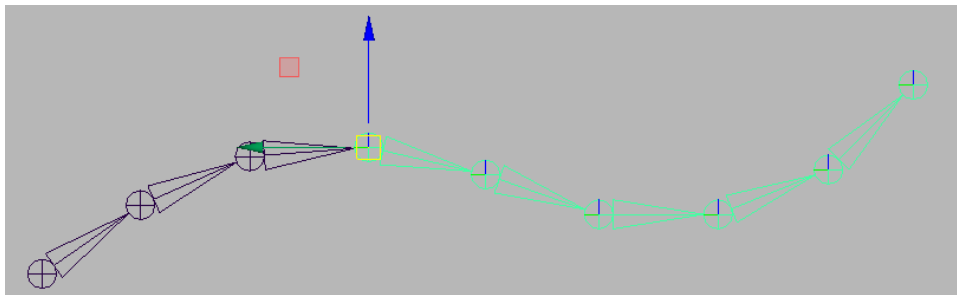
TOOL part:



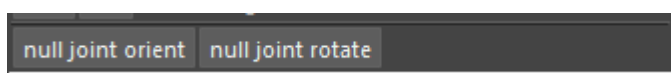
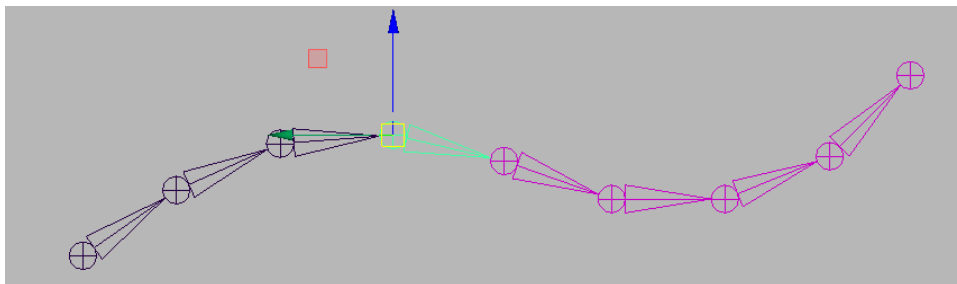
Remove gimbal rotation, Remove 360 rotation from selected



Child HL mode ON

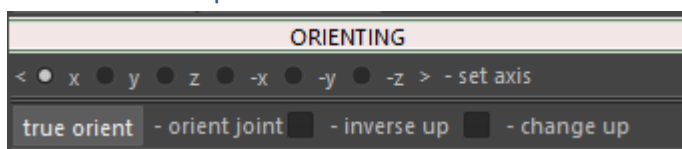


OFF



Set to zero joint orient or rotation – ignore child

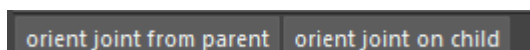
ORIENTING part:



Best orienting tool EVER!

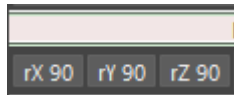
Just select joints and orient!!

<https://youtu.be/j860VDbftdk> -see for detail

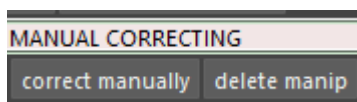


Orient joint selected Axis FROM parent or ON children direction

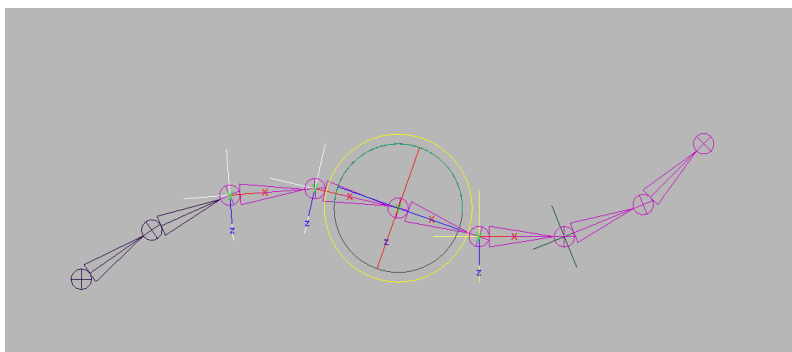
MANUAL CORRECTING part:



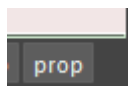
Rotate selected direction around local axis – without children direction change



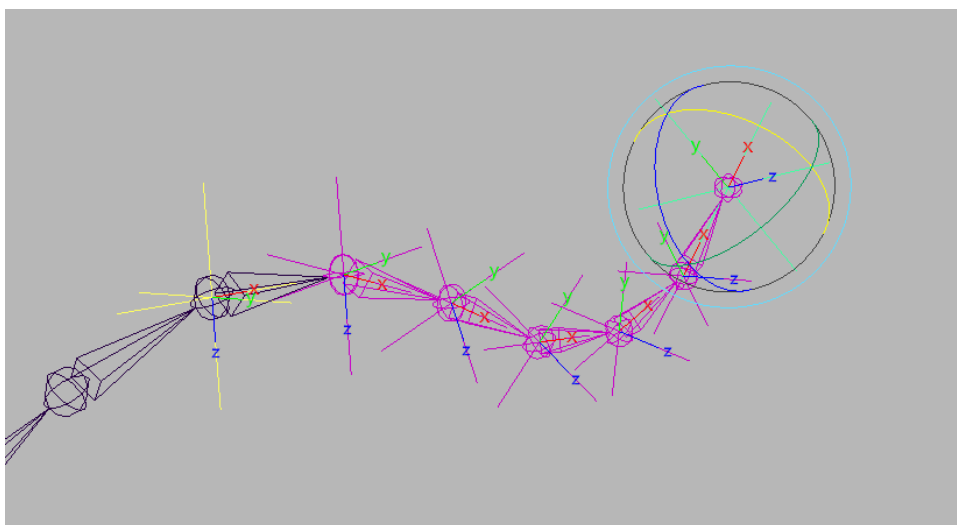
Create individually rotating system for selected joints



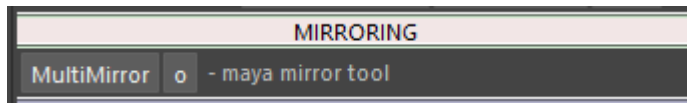
And delete it



Proportional rotating selected joints (by Manual correcting system)

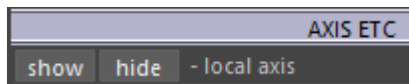


MIRRORING part:

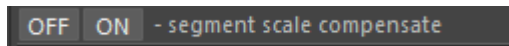


mirror selected joints with Maya options

AXIS part



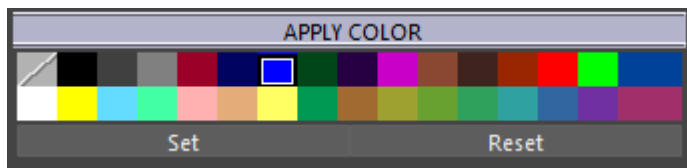
Show\Hide local axis on selected



On\Off segment scale compensate on selected joints (important for ingame skeleton)

APPLY COLOR part:

Applying selected color on selected object



NAMING tab:

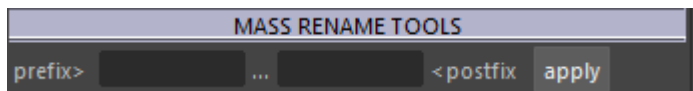
- Press names (final name replaced)
- Press POSTFIXES - (adding to final name)
- Or just type something
- And press APPLY TO SELECTED



find duplicates - finds the same names on selected

Find duplicate names from selected objects

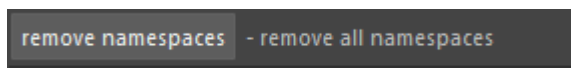
And create name_Duplicates set in Outliner



Add PREFIX and POSTFIX to selected objects



Replacing part of name selected objects



Remove all namespaces in scene excluding referenced objects

